

MOHAMED SALAH

DATA ANALYTICS | SCENARIO PLANNING | PERFORMANCE REPORTING

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PROFESSIONAL PROFILE

MSc Computer Science with Artificial Intelligence graduate (2:1) with a strong quantitative background in data analytics, predictive modelling, performance reporting and scenario planning. Experienced in preparing and validating large datasets, identifying trends, comparing actual performance against targets and translating complex analysis into clear, decision-ready insights using Python, SQL, Excel, Power Query and Power BI. Portfolio projects include an NHS emergency-department operational performance dashboard and a local-authority population and infrastructure scenario model, demonstrating experience in KPI development, variance analysis, data-quality assurance, demographic projections and long-term planning. Brings strong attention to transparent assumptions, analytical accuracy and stakeholder communication, with a genuine interest in supporting Severn Trent's water-resource planning, performance monitoring and investment decisions.

CORE SKILLS

- **Data Analysis & Scenario Planning:** Python (pandas, NumPy, scikit-learn), regression, trend analysis, feature engineering, predictive modelling, time-series analysis and scenario development.
- **BI & Performance Reporting:** Power BI, Power Query and Excel; KPI development, target and variance analysis, rolling averages, provider or area comparisons, dashboard design and recurring performance reporting.
- **Data Management & Quality:** SQL/MySQL, CSV and JSON processing, data cleansing, missing-data treatment, validation, standardisation and reproducible data-transformation processes.
- **Model Evaluation & Communication:** MAE, RMSE, MAPE and R^2 , cross-validation, sensitivity testing, interpretation of assumptions and limitations, technical reporting, presentations and stakeholder communication.

ANALYTICAL PROJECTS

NHS Emergency Department Operational Performance Dashboard | *Power BI, Power Query, SQL, Excel* | 2026

- Consolidated NHS England Monthly A&E Attendances and Emergency Admissions datasets from 2021–22 to 2025–26, standardising provider information, removing duplicate records and creating a structured analytical data model.
- Developed Power BI dashboards with DAX measures to monitor four-hour performance, target variance, year-on-year trends, rolling averages and provider rankings through interactive visualisations.
- Documented data transformations, KPI definitions and validation processes to improve transparency, reporting consistency and support recurring operational performance reporting.

Local Authority Population and Infrastructure Scenario Model | *Python, pandas, SQL, Excel, Power BI* | 2026

- Analysed ONS Subnational Population Projections to model projected population growth and identify potential future pressures on public-service infrastructure across English local authorities.
- Cleaned and validated population datasets, standardised local authority information and loaded processed records into a relational SQL database for analysis.
- Developed baseline and alternative growth scenarios, producing Power BI visualisations and evidence-based reports highlighting demographic trends, modelling assumptions and long-term infrastructure implications.

Air Quality Index Prediction and Trend Analysis | *Python, pandas, scikit-learn* | 2025

- Analysed more than 29,500 daily city-level records and 11 pollutant variables, treating missing values and structuring the dataset for reliable modelling.
- Built and compared Linear Regression, Random Forest and Gradient Boosting models to predict AQI, using RMSE, MAE and R^2 to assess accuracy and select the strongest approach.
- Interpreted pollutant patterns and model outputs, presenting clear conclusions on environmental trends, data limitations and public-health relevance.

ANALYTICAL PROJECTS (CONTINUED)

Business Data Analytics Consultancy Report | *Research and Consultancy Skills Group Project* | 2025

- Helped produce a consultancy-style report advising a small business on how data analytics could improve profitability, internal processes, automation and customer impact.
- Applied a structured evidence-review methodology to assess the credibility, relevance and recency of research before comparing analytics technologies and business use cases.
- Translated findings into practical recommendations covering data capture, storage, analysis and staff development, and contributed to the final stakeholder presentation.

Quantitative Cancer Cell Analysis Dissertation | *Fiji/ImageJ, quantitative analysis* | 2023

- Analysed thousands of microscopy images to quantify cancer-cell proliferation and support comparison across experimental conditions.
- Developed an automated counting method using image thresholding and particle analysis, improving consistency and reducing the risk of manual counting error.
- Validated the method against manually reviewed samples and reported the process, results and limitations through a structured scientific dissertation.

EDUCATION

MSc Computer Science with Artificial Intelligence (2:1) | [Keele University](#) | 2023–2025; graduated January 2026

Relevant study: Statistical Data Analytics and Databases, Applications of AI and Machine Learning, Research and Consultancy Skills, Software Engineering and Web Technologies.

BSc (Hons) Biochemistry | [University of Nottingham](#) | 2019–2023

Developed a strong scientific foundation in quantitative research, data interpretation, experimental methodology and evidence-based reporting.

EMPLOYMENT HISTORY

Parcel Delivery Driver | [Royal Mail](#) | 2025–2026 | Birmingham

- Used digital route-planning and handheld tracking systems to manage high-volume workloads and maintain accurate, auditable delivery records.
- Monitored progress against time-critical targets, reprioritising routes and tasks when volumes, access or operating conditions changed.

Quality Inspector | [WHS Plastics](#) | 2024 | Birmingham

- Inspected automotive components against technical specifications and quality standards, maintaining accurate records of checks and non-conformities.
- Identified defects and escalated findings to support corrective action, applying close attention to detail in a target-driven manufacturing environment.

Customer Service Advisor | [Halfords](#) | 2023–2024 | Redditch

- Managed customer requests, bookings and service records across several digital systems, checking information carefully and coordinating with internal teams.
- Explained service information clearly, resolved queries and escalations, and maintained structured notes to support consistent follow-up.

ADDITIONAL INFORMATION

Technical tools: Python, SQL/MySQL, Excel, Power BI, Power Query, Git/GitHub, Visual Studio and Google Colab. **Languages:** English and Arabic (fluent). **Other:** Full UK driving licence; based in Birmingham and able to work from Coventry.